

Supporting Information

Discovery, Optimization, and Biological Evaluation of 5-(2-Trifluoromethyl)phenyl)-indazoles as a Novel Class of Transient Receptor Potential A1 (TRPA1) Antagonists

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General procedure for the Synthesis of Acetyl protected Indazoles from the 2-methylaniline (Route A): 1-(5-Bromo-6-chloro-indazol-1-yl)-ethanone (7a)

N-(4-bromo-5-chloro-2-methylphenyl)acetamide

To 4-bromo-5-chloro-2-methylaniline (10 g, 45.4 mmol) in toluene (50 mL) was added acetic anhydride (4.71 mL, 49.9 mmol) and the mixture was heated to 120 °C for 1 h. The solvent was evaporated and the residue used without purification in the next step.

1-(5-Bromo-6-chloro-indazol-1-yl)-ethanone (7a)

To *N*-(4-bromo-5-chloro-2-methylphenyl)acetamide (11.91 g, 45.4 mmol) in chloroform (120 mL) was added potassium acetate (8.90 g, 91 mmol), acetic anhydride (12.84 mL, 136 mmol), 18-crown-6 (0.600 g, 2.268 mmol) and isoamyl nitrite (12.22 mL, 91 mmol) and mixture was heated to 65 °C overnight. Mixture was partitioned between water and dichloromethane. The organic phase was separated, dried over magnesium sulfate and the solvent was evaporated. Trituration with diethyl ether gave the title compound as a pale orange solid (9.2 g, 33.6 mmol, 74.1 % yield over 2 steps): ¹H NMR (400 MHz, DMSO-*d*₆) 8.45 (s, 2H), 8.40 (s, 1H), 2.70 (s, 3H).

1-(5-Bromo-6-ethyl-1H-indazol-1-yl)ethanone (7b)

4-Bromo-5-ethyl-2-methylaniline

To an ice-cooled solution of 5-ethyl-2-methyl-phenylamine (commercially available) (3.61 g, 26.7 mmol) in *N,N*-dimethylformamide (100 mL) was added *N*-bromosuccinimide (4.89 g, 27.5 mmol) portionwise. The ice-cooled mixture was stirred for 30 minutes and then allowed to warm to rt over 2 h. The reaction was quenched with water and left to stand overnight at rt. The mixture was diluted with 1:1 ethyl acetate:diethyl ether and washed with water. The organic portion was dried over sodium sulfate, filtered and concentrated *in vacuo* to give the title compound as a brown oil (4.14 g, 19.34 mmol, 72% yield): MS(ESI) *m/z* 215.9 [M+H]⁺.

1-(5-Bromo-6-ethyl-1H-indazol-1-yl)ethanone (7b)

This compound is prepared analogously to **7a** (Route A) by replacing 4-bromo-5-chloro-2-methylaniline with 4-bromo-5-ethyl-2-methylaniline: ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.37 (s, 1H), 8.25 (s, 1H), 8.16 (s, 1H), 2.86 (q, *J* = 7.5 Hz, 2H), 2.70 (s, 3H), 1.22 (t, *J* = 7.5 Hz, 3H).

1-(5-Bromo-6-isopropyl-1H-indazol-1-yl)ethanone (7c)

This compound is prepared analogously to **7b** by replacing 5-ethyl-2-methyl-phenylamine with 5-isopropyl-2-methylaniline: MS(ESI) *m/z* 281.05 [M+H]⁺.

1-(5-Bromo-6-ethoxy-1H-indazol-1-yl) ethanone (7d)

4-ethoxy-1-methyl-2-nitrobenzene

To 4-methyl-3-nitrophenol (3 g, 19.59 mmol) in *N,N*-dimethylformamide (50 mL) was added potassium carbonate (8.12 g, 58.8 mmol) and iodoethane (1.583 mL, 19.59 mmol) and mixture was stirred at rt overnight. The mixture was partitioned between water and ethyl acetate. The organic phase was separated and washed with brine, dried over magnesium sulfate and the

solvent was evaporated. The residue was used as is in next step (3.55 g, 19.59 mmol, 100 % yield).

5-ethoxy-2-methylaniline

To 4-ethoxy-1-methyl-2-nitrobenzene (3.55 g, 19.59 mmol) in ethanol (75 mL) was added ammonium formate (12.35 g, 196 mmol) and palladium on carbon (0.209 g, 1.959 mmol) and the mixture was heated to reflux for 30 minutes. The mixture was cooled, filtered through celite and the solvent was evaporated. Ethyl acetate was added to residue and the mixture was filtered. The filtrate was evaporated to leave an oil which was used without further purification in the next step (2.9 g, 19.18 mmol, 98 % yield).

1-(5-Bromo-6-ethoxy-1H-indazol-1-yl) ethanone (7d)

This compound is prepared analogously to 1-(5-bromo-6-ethyl-1H-indazol-1-yl)ethanone by replacing 5-ethyl-2-methyl-phenylamine with 5-ethoxy-2-methylaniline: ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.31 (s, 1H), 8.18 (s, 1H), 7.89 (s, 1H), 4.20 (q, *J* = 6.9 Hz, 2H), 2.70 (s, 3H), 1.41 (t, *J* = 6.9 Hz, 3H).

1-(5-Bromo-6-isopropoxy-1H-indazol-1-yl)ethanone (7e)

This compound is prepared analogously to **7d** by replacing iodoethane with 2-bromopropane: ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.30 (s, 1H), 8.15 (s, 1H), 7.90 (s, 1H), 4.75 (q, *J* = 6.0 Hz, 2H), 2.70 (s, 3H), 1.38 (d, *J* = 6.0 Hz, 6H).

1-(5-Bromo-6-trifluoromethoxy-indazol-1-yl)-ethanone (7f)

1-Methyl-2-nitro-4-(trifluoromethoxy)benzene

To 1-methyl-4-(trifluoromethoxy)benzene (10 g, 56.8 mmol) in acetic acid (12.7 mL) at 0°C was added sulphuric acid (16.95 mL, 318 mmol) followed by nitric acid (6.85 mL, 153 mmol) and the mixture was stirred for 30 min. The resulting mixture was partitioned between water and ethyl acetate. The organic phase was separated and the aqueous layer extracted with ethyl acetate. The combined organic extracts were dried over magnesium sulfate and concentrated *in vacuo*. Purification by chromatography on silica (0-100% EtOAc/iso-hexane) afforded the title product as a yellow oil (1g, 4.5 mmol, 8% yield).

2-Methyl-5-(trifluoromethoxy)aniline

To 1-methyl-2-nitro-4-(trifluoromethoxy)benzene (1 g, 4.52 mmol) in ethanol (20 mL) was added ammonium formate (0.285 g, 4.52 mmol) and palladium on carbon (0.481 g, 4.52 mmol) and mixture was heated to reflux for 30 minutes. After cooling to rt, the mixture was filtered through Celite® and the filtrate was concentrated *in vacuo* to afford the title compound as an oil (500 mg, 2.62 mmol, 58% yield): MS(ESI) *m/z* 192.08 [M+H]⁺.

1-(5-Bromo-6-trifluoromethoxy-indazol-1-yl)-ethanone (7f)

The title compound is prepared analogously to **7a** by replacing 5-chloro-2-methylaniline with 2-methyl-5-(trifluoromethoxy)aniline: ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.52 (s, 1H), 8.48 (s, 1H), 8.35 (s, 1H), 2.72 (s, 3H).

General procedure for the Synthesis of NH-Indazoles from the 2-methylaniline (Route B): 5-Bromo-6-methoxy-1H-indazole (6d)

4-bromo-5-methoxy-2-methylbenzenediazonium tetrafluoroborate

To boron trifluoride etherate (0.985 g, 6.94 mmol, 0.88 ml) in dichloromethane (10 mL) under nitrogen at -78°C was added 4-bromo-5-methoxy-2-methylaniline (1 g, 4.63 mmol) in dichloromethane (3 ml) followed by dropwise addition of tert-butylnitrite (0.66 mL, 5.5 mmol). The mixture was allowed to warm to rt and was stirred overnight. The resulting mixture was used in the next step without further purification. A sample was filtered and washed with dichloromethane and used for analysis: ^1H NMR (400MHz, $\text{DMSO}-d_6$) δ 8.37 (s, 1H), 8.22 (s, 1H), 3.99 (s, 3H), 2.65 (s, 3H).

5-Bromo-6-methoxy-1H-indazole

To the above solution of 4-bromo-5-methoxy-2-methylbenzenediazonium tetrafluoroborate in dichloromethane was added potassium acetate (861 mg, 8.77 mmol) and 18-crown-6 (58 mg, 0.219 mmol). After stirring at rt for 1 h, the mixture was filtered and the solid washed with dichloromethane. The filtrate was washed with water, brine, dried over magnesium sulfate and the solvent was removed *in vacuo*. The crude residue was purified by column chromatography on silica (0-100% EtOAc/iso-hexane) to afford the title compound (432 mg, 1.9 mmol, 43% yield): ^1H NMR (400 MHz, $\text{DMSO}-d_6$) δ 13.0 (s, 1H), 8.0 (s, 1H), 7.95 (s, 1H), 7.10 (s, 1H), 3.9 (s, 3H); MS(ESI) m/z 227.0 $[\text{M}+\text{H}]^+$.

5-Bromo-6-fluoro-1H-indazole (6b)

This compound is prepared analogously to 5-bromo-6-methoxy-1H-indazole (Route B) by replacing 4-bromo-5-methoxy-2-methylaniline with 4-bromo-5-fluoro-2-methylaniline: ^1H NMR (400 MHz, CDCl_3) δ 8.05 (s, 1H), 8.00 (d, $J = 6.4$ Hz, 1H), 7.30 (s, 1H); MS m/z 217.0 $[\text{M}+\text{H}]^+$.

5-Bromo-6-cyclopropylmethoxy-1H-indazole (6e)

4-(Cyclopropylmethoxy)-1-methyl-2-nitrobenzene

4-Methyl-3-nitrophenol (2 g, 13.06 mmol) in dry N,N-dimethylformamide (50 mL) was treated with cesium carbonate (12.77 g, 39.2 mmol) followed by (bromomethyl)cyclopropane (1.89 mL, 19.59 mmol) and the mixture was heated to 80°C for 30 minutes. After cooling to rt overnight, the reaction mixture was filtered and concentrated *in vacuo*. The crude residue was dissolved in ethyl acetate and washed with 1M sodium hydroxide, water, brine, dried over magnesium sulfate and concentrated *in vacuo* to give an oil. The oil was dissolved in ethyl acetate and washed with 0.5M lithium chloride, dried over magnesium sulfate and concentrated *in vacuo* to afford the title compound (2.7 g, 13.0 mmol, 100% yield): ^1H NMR (400MHz, CDCl_3) δ 7.50 (d, $J = 2.5$ Hz, 1H), 7.22 (d, $J = 8.5$ Hz, 1H), 7.09 (dd, $J = 8.5, 2.5$ Hz, 1H), 3.85 (d, $J = 6.9$ Hz, 2H), 2.52 (s, 3H), 1.29 (m, 1H), 0.69 (m, 2H), 0.39 (m, 2H).

5-(Cyclopropylmethoxy)-2-methylaniline

4-(Cyclopropylmethoxy)-1-methyl-2-nitrobenzene (2.47 g, 11.9 mmol) in ethanol (248 mL) and water (50 ml) was treated with ammonium chloride (1.27 g, 23.8 mmol) and heated to 70°C .

Iron powder (3.33 g, 59.6 mmol) was added and the mixture was stirred for 2 h at 70°C. The hot mixture was filtered through Celite® and washed with ethyl acetate. The filtrate was concentrated *in vacuo* and the crude product was purified by chromatography on silica (0-50% EtOAc/iso-hexane) to afford the title compound (1.74 g, 9.83 mmol, 83% yield): ¹H NMR (400MHz, CDCl₃) δ 6.95 (d, *J* = 8.9 Hz, 1H), 6.30 (m, 2H), 3.75 (d, *J* = 6.9 Hz, 2H), 3.61 (br s, 2H), 2.11 (s, 3H), 1.29 (m, 1H), 0.65 (m, 2H), 0.35 (m, 2H); MS(ESI) *m/z* 178.2 [M+H]⁺.

4-Bromo-5-(cyclopropylmethoxy)-2-methylaniline

A solution of 5-(cyclopropylmethoxy)-2-methylaniline (1.576 g, 8.89 mmol) in N,N-dimethylformamide (25 mL) was cooled in an ice bath and treated with N-bromosuccinimide (1.583 g, 8.89 mmol). The resulting mixture was allowed to warm to room temperature and left to stir for 2 hours. The mixture was partitioned between water and ethyl acetate. The organic phase was separated, dried over magnesium sulfate and concentrated *in vacuo*. The residue was purified by chromatography on silica (0-50% EtOAc/iso-hexane) to afford the title compound (1.56 g, 6.08 mmol, 68% yield): ¹H NMR (400MHz, CDCl₃) δ 7.19 (s, 1H), 6.30 (s, 1H), 3.82 (d, *J* = 6.7 Hz, 1H), 3.69 (br s, 2H), 2.09 (s, 3H), 1.31 (m, 1H), 0.64 (m, 2H), 0.39 (m, 2H); MS(ESI) *m/z* 256.2 [M+H]⁺.

5-Bromo-6-cyclopropylmethoxy-1H-indazole

This compound is prepared analogously to 5-Bromo-6-methoxy-1H-indazole by replacing 4-bromo-5-methoxy-2-methylaniline with 4-bromo-5-(cyclopropylmethoxy)-2-methylaniline: ¹H NMR (400MHz, CDCl₃) δ 7.95 (s, 2H), 6.90 (s, 1H), 3.95 (d, *J* = 6.6 Hz, 2H), 1.39 (m, 1H), 0.70 (m, 2H), 0.46 (m, 2H); MS(ESI) *m/z* 269.2 [M+H]⁺.

5-Bromo-6-(difluoromethoxy)-1H-indazole (6f)

The title compound is prepared analogously to 5-bromo-6-cyclopropylmethoxy-1H-indazole by replacing (bromomethyl)cyclopropane with sodium 2-chloro-2,2-difluoroacetate: MS(ESI) *m/z* 263.3 [M+H]⁺.

5-Bromo-1H-indazole-6-carbonitrile (6g)

The title compound is prepared analogously to 5-bromo-6-cyclopropylmethoxy-1H-indazole by replacing 5-(cyclopropylmethoxy)-2-methylaniline with 3-amino-4-methylbenzonitrile: ¹H NMR (400 MHz, DMSO-*d*₆) δ 13.82 (s, 1H), 8.32 (s, 2H), 8.22 (s, 1H); MS(ESI) *m/z* 263.3 [M+CH₃CN]⁺.

Methyl 5-bromo-1H-indazole-6-carboxylate (6h)

The title compound is prepared analogously to 5-bromo-6-cyclopropylmethoxy-1H-indazole by replacing 5-(cyclopropylmethoxy)-2-methylaniline with methyl 3-amino-4-methylbenzoate: ¹H NMR (400 MHz, DMSO-*d*₆) δ 13.55 (s, 1H), 8.20 (s, 1H), 8.16 (s, 1H), 7.97 (s, 1H), 3.90 (s, 3H); MS(ESI) *m/z* 257.0 [M+H]⁺.

5-Bromo-4-fluoro-1H-indazole (6i)

This compound is prepared analogously to 5-bromo-6-methoxy-1H-indazole by replacing 4-bromo-5-methoxy-2-methylaniline with 4-bromo-3-fluoro-2-methylaniline: ¹H NMR (400 MHz,

DMSO-*d*₆) δ 8.22 (t, J = 1.3 Hz, 1H), 7.52 (dd, J = 8.8, 6.4 Hz, 1H), 7.38 (dd, J = 8.7, 1.1 Hz, 1H); MS(ESI) m/z 214.0 [M+H]⁺.

5-Bromo-7-methyl-1H-indazole (6j)

This compound is prepared analogously to 5-bromo-6-methoxy-1H-indazole by replacing 4-bromo-5-methoxy-2-methylaniline with 4-bromo-2,6-dimethylaniline: ¹H NMR (400 MHz, DMSO-*d*₆) δ 8.03 (s, 1H), 7.78 (dd, J = 1.8, 0.8 Hz, 1H), 7.25 (dt, J = 2.0, 1.1 Hz, 1H), 2.52 – 2.49 (m, 3H); MS(ESI) m/z 211.0 [M+H]⁺.

The following compounds were prepared following the procedure given for Compound 17 using the appropriate boronic acid and 5-bromoindazole:

5-(2-Chlorophenyl)-1H-indazole (4)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.13 (s, 1H), 7.79 (s, 1H), 7.62 (d, J = 8.4 Hz, 2H), 7.58 (d, J = 8.3 Hz, 1H), 7.50-7.36 (m, 4H); MS(ESI) m/z 229.0 [M+H]⁺.

5-Phenylindazole (11)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.10 (s, 1H), 7.97 (s, 1H), 7.62 (ddd, J = 10.3, 5.9, 2.3 Hz, 4H), 7.41 (t, J = 7.7 Hz, 2H), 7.33 – 7.24 (m, 1H); MS(ESI) m/z 195.0 [M+H]⁺.

5-(2-Fluorophenyl)-1H-indazole (12)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.13 (s, 1H), 7.79 (s, 1H), 7.61 (d, J = 8.4 Hz, 2H), 7.58 (d, J = 8.3 Hz, 1H), 7.47-7.37 (m, 4H); MS(ESI) m/z 231.1 [M+H]⁺.

5-(2-(Difluoromethyl)phenyl)-1H-indazole (18)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.18 (s, 1H), 7.81-7.73 (m, 2H), 7.69-7.56 (m, 4H), 7.47 (d, J = 7.0 Hz, 1H), 7.36 (d, J = 8.8 Hz, 1H); MS(ESI) m/z 245.1 [M+H]⁺.

5-(2-(Trifluoromethoxy)phenyl)-1H-indazole (19)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.14 (s, 1H), 7.84 (s, 1H), 7.64-7.55 (m, 2H), 7.53-7.43 (m, 4H); MS(ESI) m/z 279.1 [M+H]⁺.

5-(5-Fluoro-2-(trifluoromethyl)phenyl)-1H-indazole (23)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.14 (s, 1H), 7.92 (dd, J = 8.4, 5.7 Hz, 1H), 7.73 (s, 1H), 7.60 (d, J = 8.5 Hz, 1H), 7.49-7.43 (m, 1H), 7.36-7.27 (m, 2H); MS(ESI) m/z 281.1 [M+H]⁺.

5-(6-Fluoro-2-(trifluoromethyl)phenyl)-1H-indazole (24)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.13 (s, 1H), 7.74-7.58 (m, 3H), 7.35-7.29 (m, 1H), 7.28-7.21 (m, 1H), 7.17-7.08 (m, 1H); MS(ESI) m/z 281.1 [M+H]⁺.

4-Fluoro-5-(2-trifluoromethyl-phenyl)-1H-indazole (25)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.22 (s, 1H), 7.88 (d, J = 8.0 Hz, 1H), 7.76 (t, J = 7.6 Hz, 1H), 7.67 (t, J = 8.0 Hz, 1H), 7.49 (d, J = 7.0 Hz, 1H), 7.44 (d, J = 8.7 Hz, 1H), 7.23 (t, J = 7.5 Hz, 1H); MS(ESI) m/z 281.0 [M+H]⁺.

3-Methyl-5-(2-(trifluoromethyl)phenyl)-1H-indazole (27)

¹H NMR (400 MHz, DMSO-*d*₆) δ 7.81 (d, *J* = 7.4 Hz, 1H), 7.71-7.68 (m, 1H), 7.61-7.55 (m, 2H), 7.53-7.48 (m, 1H), 7.42 (d, *J* = 7.8 Hz, 2H), 7.29-7.22 (m, 1H), 2.46 (s, 3H); MS(ESI) *m/z* 277.1 [M+H]⁺.

7-Methyl-5-(2-(trifluoromethyl)phenyl)-1H-indazole (28)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.10 (s, 1H), 7.82 (d, 1H, *J* = 8.1 Hz), 7.70 (t, 1H, *J* = 7.6 Hz), 7.60 (t, 1H, *J* = 7.6 Hz), 7.49 (s, 1H), 7.43 (d, 1H, *J* = 6.8 Hz), 7.06 (s, 1H), 2.55 (s, 3H); MS(ESI) *m/z* 277.0 [M+H]⁺.

The following compounds were prepared following the procedure given for Compound 31 using the appropriate boronic acid and 5-bromoindazole:

5-o-Tolyl-1H-indazole (13)

¹H NMR (400 MHz, DMSO-*d*₆) δ 13.13 (s, 1H), 8.09 (s, 1H), 7.68 (s, 1H), 7.58 (d, *J* = 8.5 Hz, 1H), 7.33 (m, 2H), 7.24 (m, 3H), 2.23 (s, 3H); MS(ESI) *m/z* 209.21 [M+H]⁺.

6-Fluoro-5-(2-(trifluoromethyl)phenyl)-1H-indazole (26)

Mp 158-163°C; ¹H NMR (400MHz, DMSO-*d*₆) δ 13.24 (s, 1H), 8.12 (s, 1H), 7.86 (d, *J* = 7.6 Hz, 1H), 7.75 (dd, *J* = 7.5, 7.5 Hz, 1H), 7.70 (d, *J* = 7.2 Hz, 1H), 7.67 (dd, *J* = 7.8, 7.8 Hz, 1H), 7.47 (d, *J* = 7.6 Hz, 1H), 7.43 (d, *J* = 10.0 Hz, 1H); ¹³C NMR (125MHz, DMSO-*d*₆) δ 158.3 (d, *J* = 235.9 Hz), 139.6 (d, *J* = 12.8Hz), 134.6 (q, *J* = 2.0 Hz), 134.0, 132.9, 132.2, 128.7, 128.2 (q, *J* = 29.1 Hz), 126.0 (q, *J* = 5.0 Hz), 124.0 (q, *J* = 246.7 Hz), 122.8 (d, *J* = 4.3 Hz), 120.9, 119.2, 95.4 (d, *J* = 27.3 Hz); ¹⁹F NMR (376 MHz, DMSO-*d*₆) δ -57.60, -116.31; EA: Calc for C₁₄H₈F₄N₂: C, 60.01; H, 2.88; N, 10.00: Found: C, 59.28; H, 3.08, N, 9.95; MS(ESI) *m/z* 281.4 [M+H]⁺.

6-Methoxy-5-(2-(trifluoromethyl)phenyl)-1H-indazole (36)

Mp 155-162°C; ¹H NMR (400 MHz, DMSO-*d*₆) δ 12.90 (s, 1H), 7.95 (s, 1H), 7.79 (d, *J* = 7.9 Hz, 1H), 7.68 (dd, *J* = 7.9; 7.9 Hz, 1H), 7.59 (dd, *J* = 7.9; 7.9 Hz, 1H), 7.45 (s, 1H), 7.33 (d, *J* = 7.9 Hz, 1H), 7.01 (s, 1H), 3.72 (s, 3H). ¹³C NMR (125MHz, DMSO-*d*₆) δ 156.3, 140.7, 137.7, 133.6, 132.9, 131.8, 128.2 (q, *J* = 28.6 Hz), 127.7, 125.8 (q, *J* = 5.1 Hz), 124.2 (q, *J* = 272.9 Hz), 123.6, 121.5, 116.5, 90.1, 55.4; ¹⁹F NMR (400 MHz, DMSO-*d*₆) δ -57.63; EA: Calc for C₁₅H₁₁F₃N₂O: C, 61.65; H, 3.79; N, 9.58: Found: C, 61.32; H, 3.83, N, 9.54. HRMS: Calc for C₁₅H₁₂F₃N₂O: 293.0902; Found: 293.0888.

6-(Cyclopropyl methoxy)-5-(2-(trifluoromethyl)phenyl)-1H-indazole (39)

¹H NMR (400MHz, CDCl₃) δ 7.82 (s, 1H), 7.58 (d, *J* = 7.8 Hz, 1H), 7.35 (m, 3H), 7.18 (d, *J* = 7.5 Hz, 1H), 6.71 (s, 1H), 3.65 (m, 2H), 0.90 (m, 1H), 0.28 (m, 2H), 0.00 (m, 2H); MS(ESI) *m/z* 333.5 [M+H]⁺.

6-(Difluoromethoxy)-5-(2-(trifluoromethyl) phenyl)-1H-indazole (42)

¹H NMR (400 MHz, DMSO-*d*₆) δ 13.2 (s, 1H), 8.10 (s, 1H), 7.85 (d, *J* = 7.6 Hz, 1H), 7.75 (dd, *J* = 7.3; 7.3 Hz, 1H), 7.65 (m, 2H), 7.40 (m, 1H), 7.35 (s, 1H), 7.20 (t, *J* = 7.4 Hz, 1H); MS(ESI) *m/z* 329.4 [M+H]⁺.

5-(2-(Trifluoromethyl) phenyl)-1H-indazole-6-carbonitrile (43)

¹H NMR (400 MHz, CDCl₃) δ 8.22 (s, 1H), 7.99 (s, 1H), 7.83 (m, 2H), 7.64 (m, 2H), 7.42 (s, 1H); MS(ESI) *m/z* 288.0 [M+H]⁺.

The following compounds were prepared following the procedure given for Compound 15 using the appropriate aryl bromide:

5-(2-Ethyl-phenyl)-1H-indazole (14)

¹H NMR (400 MHz, DMSO-*d*₆) δ 13.1 (s, 1H), 8.09 (s, 1H), 7.65 (s, 1H), 7.58 (d, *J* = 8.5 Hz, 1H), 7.36-7.18 (m, 6H), 2.56 (q, *J* = 7.5 Hz, 2H), 1.02 (t, *J* = 7.5 Hz, 3H); MS(ESI) *m/z* 223.09 [M+H]⁺.

5-(2-tert-Butyl-phenyl)-1H-indazole (16)

¹H NMR (400 MHz, DMSO-*d*₆) δ 13.08 (s, 1H), 8.08 (s, 1H), 7.58 (s, 1H), 7.52 (m, 2H), 7.31 (t, *J* = 8.0 Hz, 1H), 7.20 (m, 2H), 6.97 (d, *J* = 7.4 Hz, 1H), 1.13 (s, 9H); MS(ESI) *m/z* 251.18 [M+H]⁺.

2-(1H-Indazol-5-yl)benzonitrile (20)

¹H NMR (400 MHz, DMSO-*d*₆) δ 13.20 (s, 1H), 8.20 (s, 1H), 7.98 (s, 1H), 7.95 (d, *J* = 7.8 Hz, 1H), 7.79 (m, 1H), 7.68 (m, 2H), 7.55 (m, 2H); MS(ESI) *m/z* 220.19 [M+H]⁺.

5-(3-Fluoro-2-trifluoromethyl-phenyl)-1H-indazole (21)

¹H NMR (400 MHz, DMSO-*d*₆) δ 13.20 (s, 1H), 8.13 (s, 1H), 7.76 (m, 1H), 7.70 (s, 1H), 7.60 (d, *J* = 8.5 Hz, 1H), 7.55 (m, 1H), 7.30 (s, 1H), 7.28 (s, 1H); MS(ESI) *m/z* 281.27 [M+H]⁺.

5-(4-Fluoro-2-trifluoromethyl-phenyl)-1H-indazole (22)

¹H NMR (400 MHz, DMSO-*d*₆) δ 8.13 (s, 1H), 7.74 (dd, *J* = 9.4, 2.7 Hz, 1H), 7.69 (s, 1H), 7.67 – 7.56 (m, 2H), 7.52 (dd, *J* = 8.6, 5.7 Hz, 1H), 7.28 (dd, *J* = 8.6, 1.5 Hz, 1H); MS(ESI) *m/z* 281.1 [M+H]⁺.

The following compounds were prepared following the procedure given for Compound 30 using the appropriate aryl boronic acid and 5-bromoindazole:

6-Ethyl-5-(2-trifluoromethyl-phenyl)-1H-indazole (32)

¹H NMR (400 MHz, CDCl₃) δ 10.0 (s, 1H), 7.97 (s, 1H), 7.70 (d, *J* = 7.8 Hz, 1H), 7.49 (m, 1H), 7.46-7.36 (m, 2H), 7.34 (s, 1H), 7.22 (d, *J* = 7.4 Hz, 1H), 2.46-2.26 (m, 2H), 1.06 (t, *J* = 7.5 Hz, 3H); MS(ESI) *m/z* 291.23 [M+H]⁺.

6-Isopropyl-5-(2-trifluoromethyl-phenyl)-1H-indazole (33)

¹H NMR (400 MHz, CDCl₃) δ 9.70 (br s, 1H), 8.12 (s, 1H), 7.81 (d, *J* = 7.7 Hz, 1H), 7.62-7.48 (m, 4H), 7.32 (d, *J* = 7.2 Hz, 1H), 2.73-2.64 (m, 1H), 1.24 (d, *J* = 6.7 Hz, 3H), 1.15 (d, *J* = 6.7 Hz, 3H); MS(ESI) *m/z* 305.20 [M+H]⁺.

6-Ethoxy-5-(2-trifluoromethyl-phenyl)-1H-indazole (37)

¹H NMR (400 MHz, DMSO-*d*₆) δ 12.90 (s, 1H), 7.95 (s, 1H), 7.79 (d, *J* = 7.8 Hz, 1H), 7.68 (dd, *J* = 7.5; 7.5 Hz, 1H), 7.59 (dd, *J* = 7.6; 7.6 Hz, 1H), 7.45 (s, 1H), 7.33 (d, *J* = 7.6 Hz, 1H), 7.01 (s, 1H), 4.02 (m, 2H), 1.13 (t, *J* = 6.9 Hz, 3H); MS(ESI) *m/z* 307.3 [M+H]⁺.

6-Isopropoxy-5-(2-trifluoromethyl-phenyl)-1H-indazole (38)

¹H NMR (400 MHz, DMSO-*d*₆) δ 12.90 (s, 1H), 7.95 (s, 1H), 7.79 (d, *J* = 7.8 Hz, 1H), 7.68 (dd, *J* = 7.5; 7.5 Hz, 1H), 7.57 (dd, *J* = 7.5; 7.5 Hz, 1H), 7.44 (s, 1H), 7.33 (d, *J* = 7.5 Hz, 1H), 7.01 (s, 1H), 4.59 (sept, *J* = 6.0 Hz, 1H), 1.12 (d, *J* = 6.0 Hz, 6H); MS(ESI) *m/z* 321.3 [M+H]⁺.

6-Trifluoromethoxy-5-(2-trifluoromethyl-phenyl)-1H-indazole (41)

¹H NMR (400 MHz, DMSO-*d*₆) δ 13.35 (s, 1H), 8.19 (s, 1H), 7.86 (d, *J* = 8.0 Hz, 1H), 7.79 (s, 1H), 7.75 (dd, *J* = 7.7, 7.7 Hz, 1H), 7.68 (dd, *J* = 7.7, 7.7 Hz, 1H), 7.60 (s, 1H), 7.41 (d, *J* = 7.6 Hz, 1H); MS(ESI) *m/z* 374.3 [M+H]⁺.